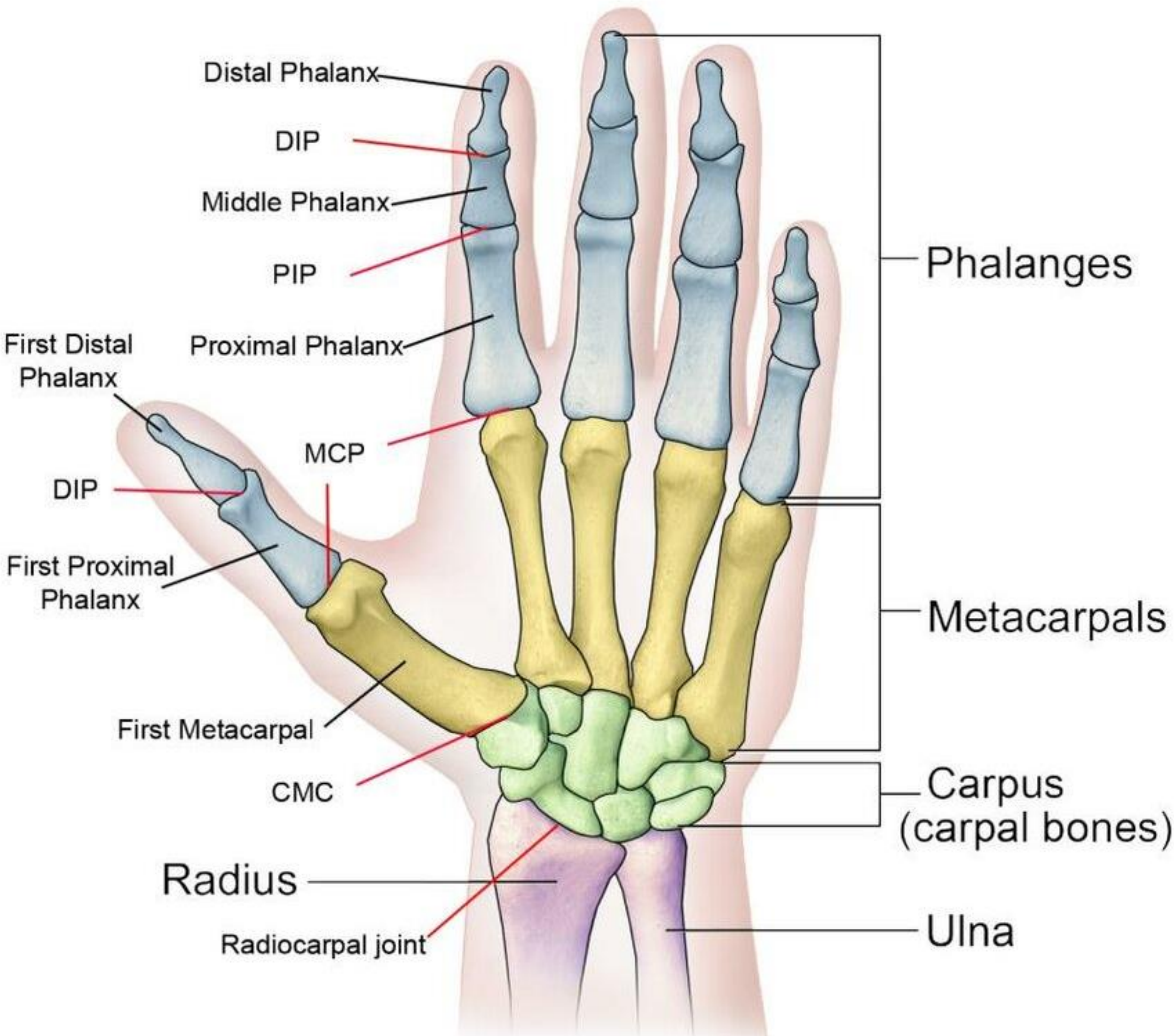
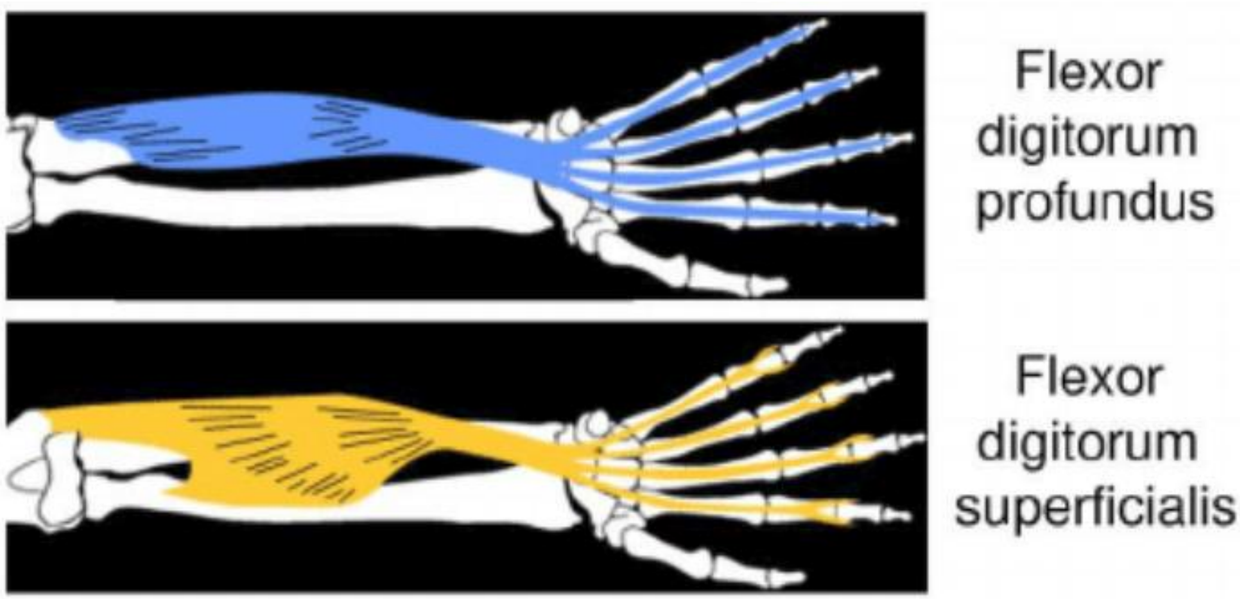
Open Grips – Isolated Frictionless Ergonomic Grip Training for FDP and FDS

Version 1.0



1. Overview:

The idea to train FDS and FDP separately using frictionless ergonomic devices to improve climbing performance was first put forward by David Quinn (mobeta) out of Halifax, NS, Canada. The idea being that most climbers will become dominant in one over the other muscle group, thereby not achieving there maximal strength potential. Finger strength on the wall will almost always use a combination of both (as well as other) muscle groups, so by isolating these two in training (as much as possible) it should be probable to see gains in performance strength. The ergonomic side should allow for maximum force exertion without stressing the finger pulleys or causing other types of finger pain like synovitis. Frictionless implementation ensures that active engagment and not passive grip is being used throught the entire exercise (skin friction vs active muscle engagement). While an oversimplification, the FDP flexes the DIP joint, and the FDS flexes the PIP joint.



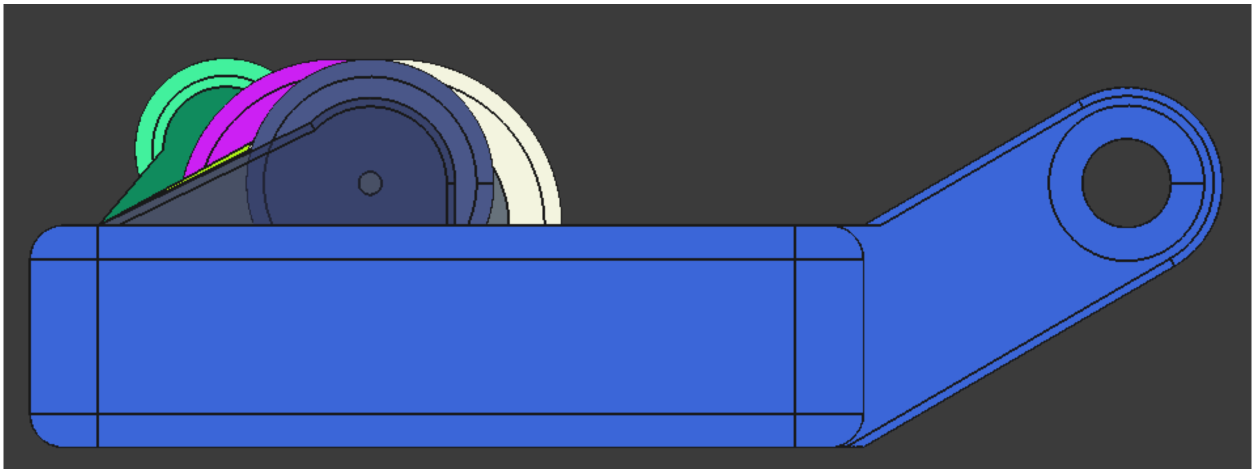
The Open Grips project aims to be a free and open source version for the community to use, experiment, and hopefully improve. It make no claims of having an optimized solution, but does aim for continuous improvement. If you want the original and professionally produced version of these training tools head on over the handofgod shop.

This project uses FreeCad for parametric modeling, and 3D printing for manufacturing.

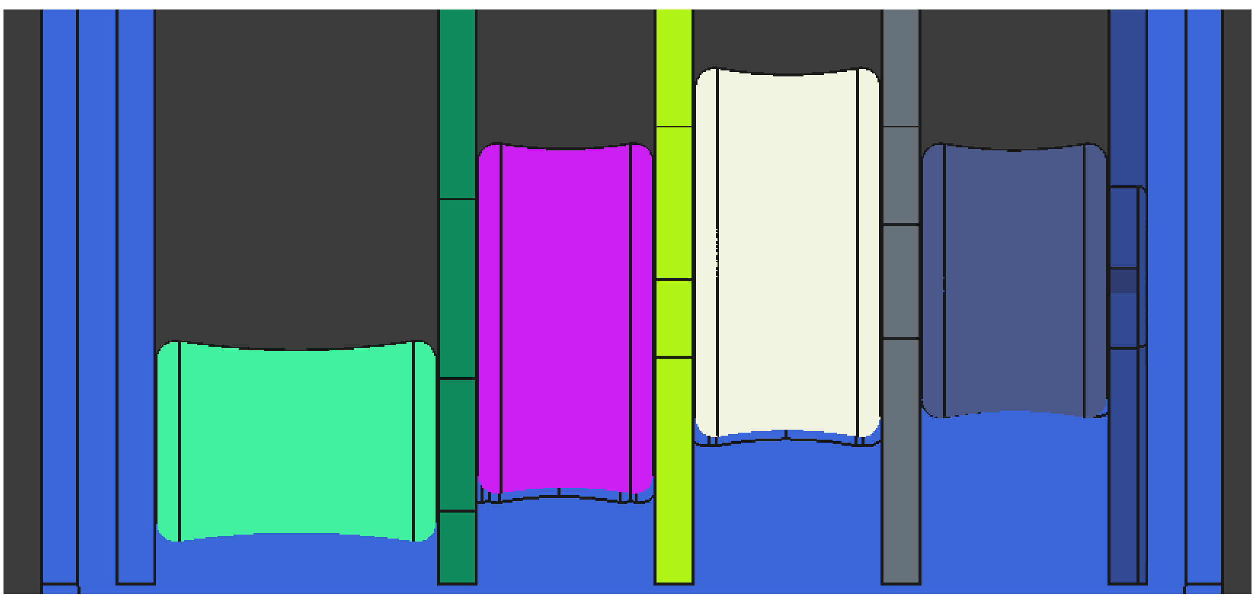
2. FDS: Prometheus Gripper

2.1. Gripper Design Concept

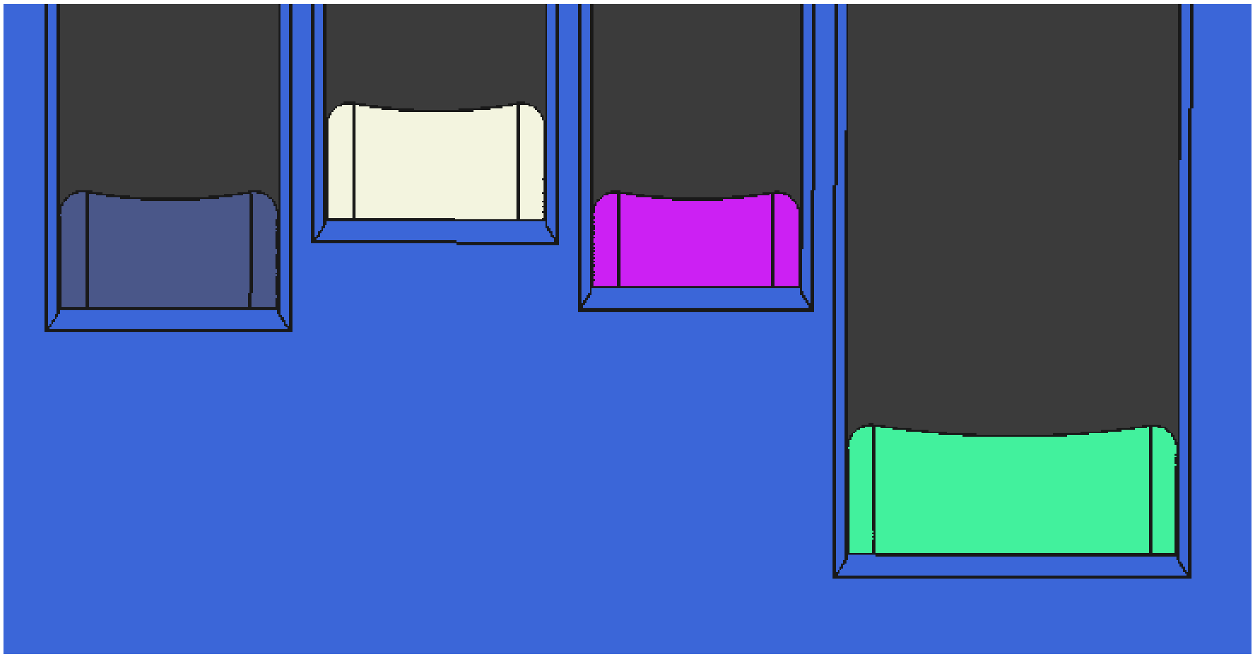
The Prometheus gripper targets the FDS by having free spinning rollers that custom fit into each finger, and parallel on the outer tangent as seen below.



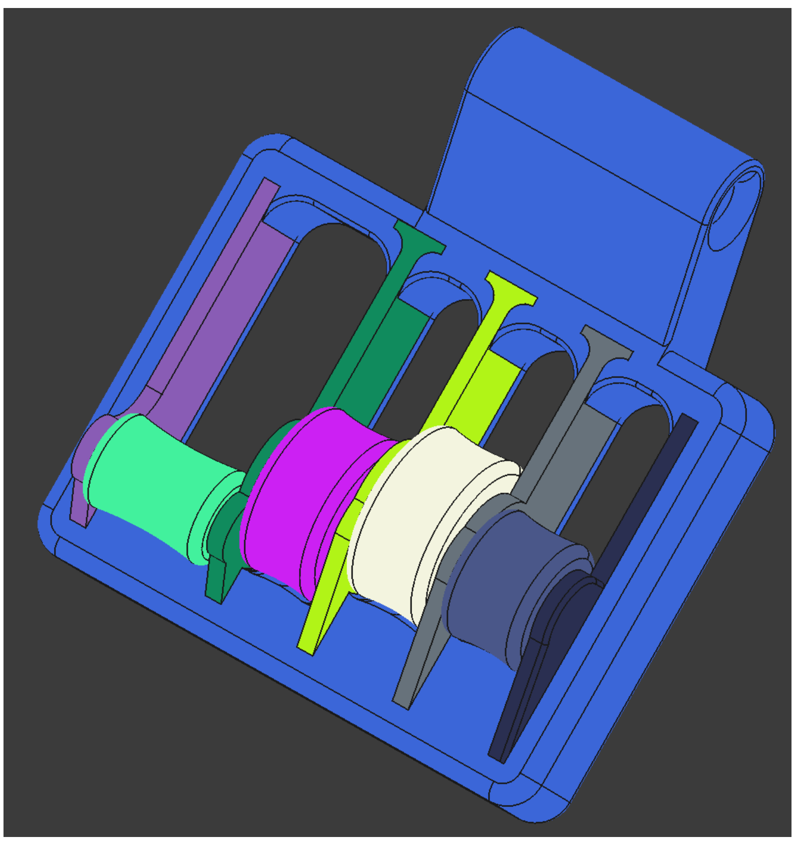
The top of each roller is offset to fit each finger height like an unleveled edge.



On the back the blockers rise so that when the tip of the fingers touch the blockers the DIP joint is at the apex of the roller. The rollers are sized so that the Proximal Phalanx is vertical.

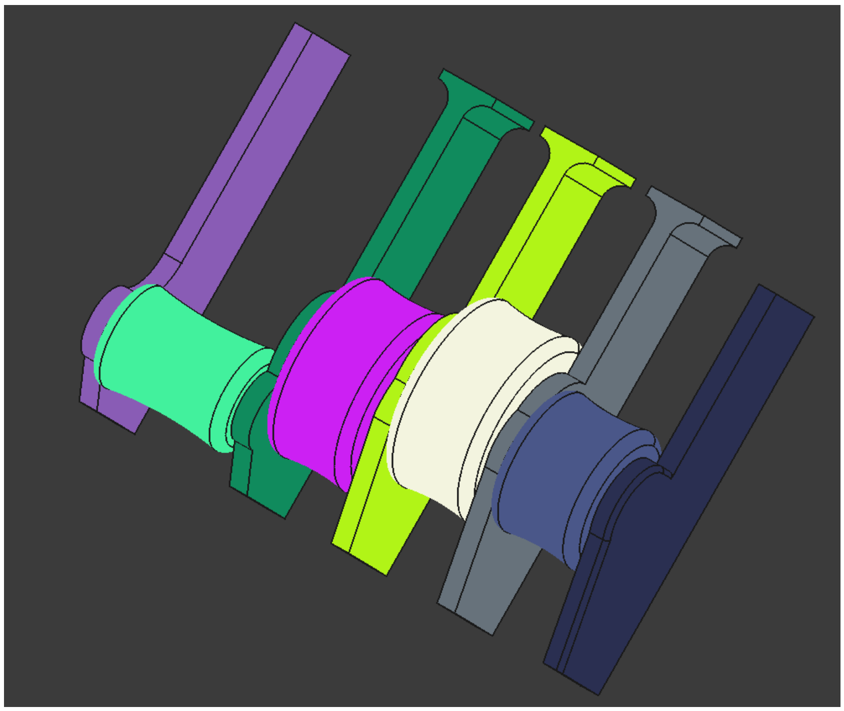


The anchor is aligned vertically with the center of the index roller, and horizontally with the wall between the pinky/ring fingers, and halfway through the index roller. This way the gripper stays level when pulling. A loose rope loop should be used through the anchor hole.



2.2. Gripper Assembly

Walls roller and frame are all printed separately then walls and rollers are assembled as seen below and then lowered into the frame. A pole for the rollers should be of metal. A 3mm nail was cut and used in the prototype. Make sure not to make the pole to long or it will puncture the walls on assembly. If printed without scaling down the walls should fit tightly. They can be tamped down by hammer with a block of wood across the tops of the walls.



2.3. Measurement Variables

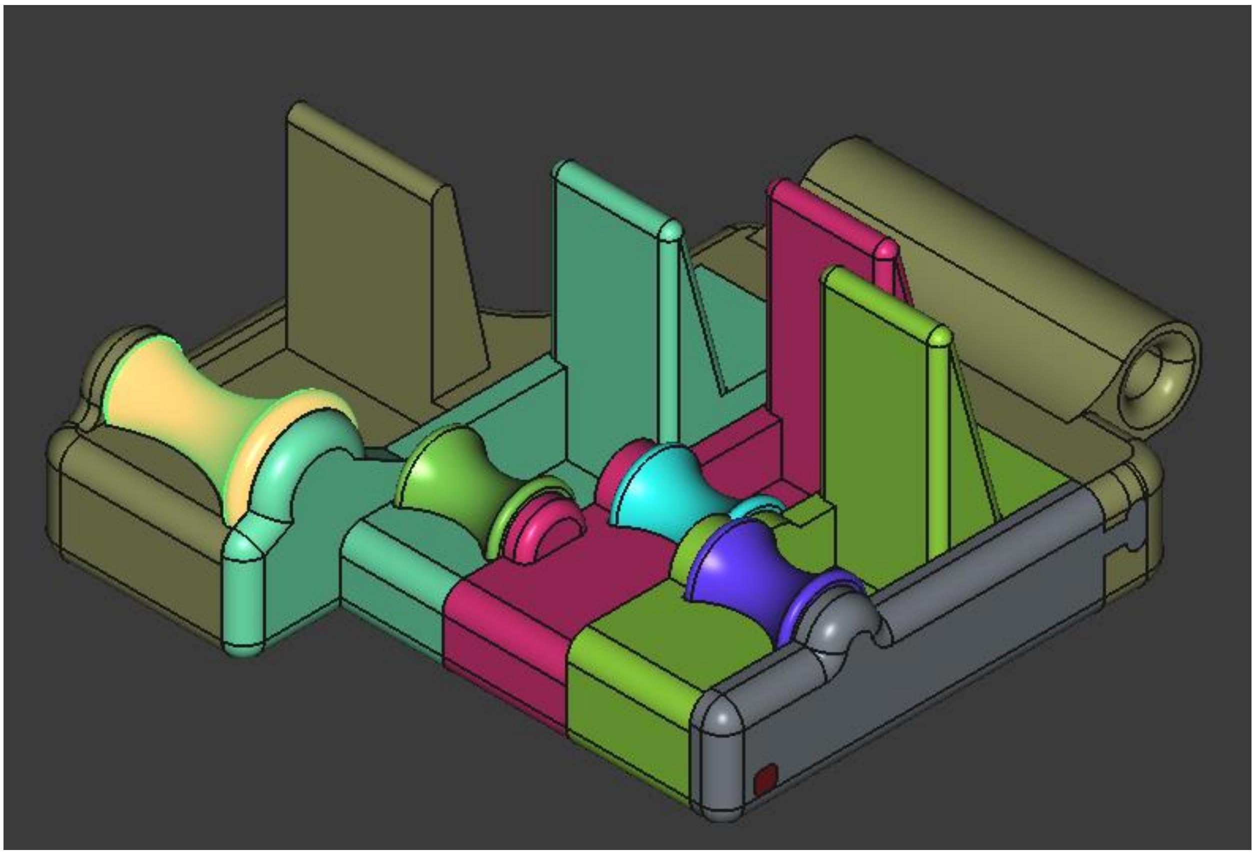
Four variables for each finger replace X with (I=index, M=middle, R=Ring, P=Pinky)

Frame_Height	Inside Height of the frame, adjust if not enough space above fingers
Pole_Width	Diameter of the metal roller poles
Wall_Width	Width of internal wall. Best not to change, 4mm seems very strong.
X_Height	Height from the bottom of the pinky to the bottom of the finger. P_Height is always zero.
X_Diameter	Diameter of each fingers roller
X_Width	Width of each fingers roller
X_Blocker	Height from the center of the roller to the top of the fingers blocker. For the Pinky this is below the center of the roller instead.

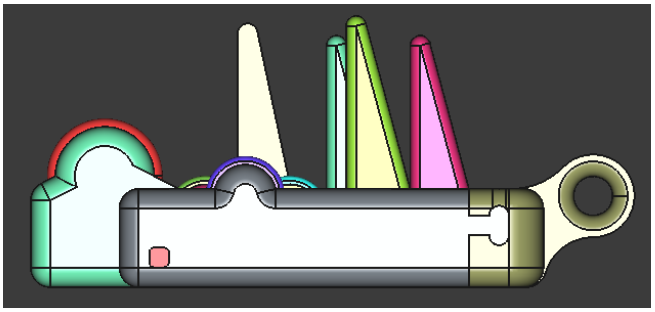
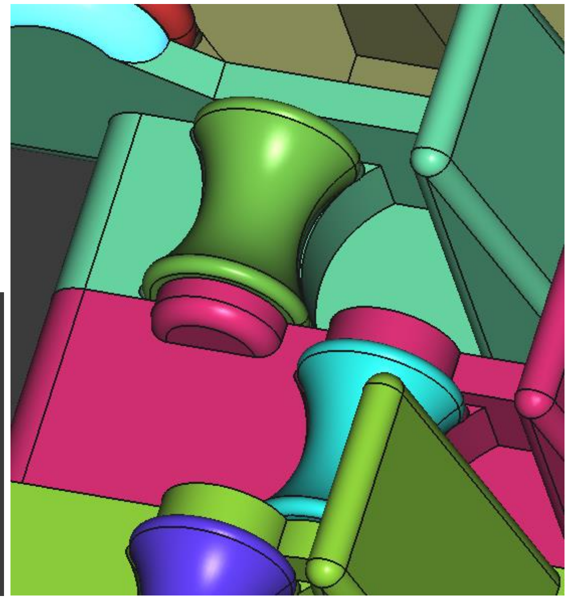
3. Design – FDP: Sisyphus Gripper

3.1. Gripper Structure

The Sisyphus gripper targets the FDP by placing all the force on the end of the distal phalanx in a claw like grip. Imagine you form your hand to hold a thin jug but only the tips of your fingers are making contact. Your PIP and DIP joints are flexed about 55 Deg each and proximal phalanx is once again vertical. For this to be the case for each finger the height and depth of each roller must be set.



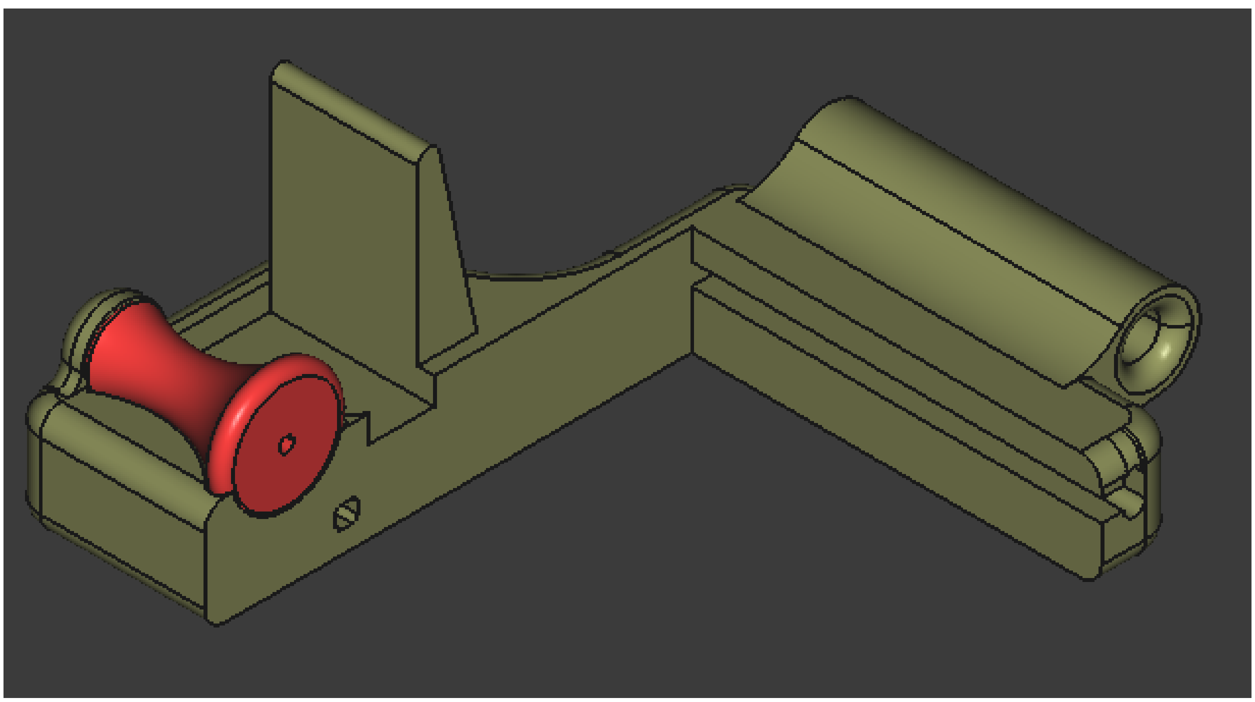
The blockers above the rollers should be touched by the DIP joint and keep the grip from becoming a crimp. The rollers have a ledge in front so roughly half a pad is making contact, quarter on the roller, quarter on the ledge.

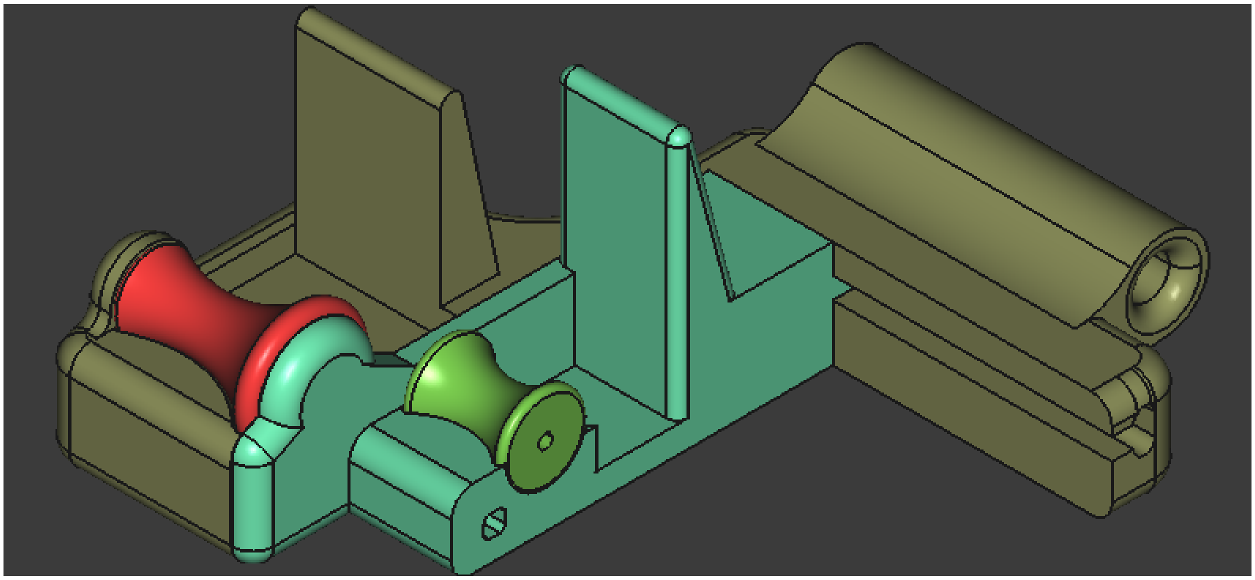
 

The anchor is designed to align like the prometheus.

3.2. Gripper Assembly

Place one roller at time, insert the pole, and then slide in the next section on the rail system and continue. Finally add the last wall and then the securing post through the hole in the side. All sections were printed without scaling down except for the post which needed to be bout 0.2mm thinner. Hammer section together gently with a hammer.





3.3. Measurement Variables

Five variables for each finger replace X with (I=index, M=middle, R=Ring, P=Pinky). Change these as needed in VarSet.

Blocker_Height	Height of blocker, shouldn't need to change
Pole_Diameter	Diameter of the metal roller poles
Wall_Width	Width of internal wall. Best not to change, 4mm seems very strong.
X_Height	Height from the bottom of the pinky to the bottom of the finger. P_Height is always zero.
X_Diameter	Diameter of each fingers roller
X_Width	Width of each fingers roller
X_To_Blocker	Height from the bottom of finger to the bottom of the blocker
X_Depth	How farther in the tip of the finger is when in the claw grip from the pinky. P_Depth is always zero.

4. Printer Settings

- **Filament**: PETG-HF
- **Layer Height**: 0.2mm
- **Nozzle Size**: 0.4mm
- **Wall Loops**: 5
- **Infill**: cubic 35%
- **Supports**: Normal (for frame only)
- **Orientation**: Rollers on their side. For the Prometheus print everything as modeled. For the Sisyphus print frame and wall on their side with holes pointing up so no supports go inside.